



MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examinations 2015
Higher 2

CANDIDATE
NAME

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CIVICS
GROUP

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INDEX
NUMBER

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H2 BIOLOGY

9648/02

Paper 2 Core Paper

17 September 2015

2 hours

Additional Materials: Answer papers

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid/tape.

Section A

Answer **all** questions in the spaces provided on the question paper.

Section B

Answer **one** question on the answer paper provided.

At the end of the examination,

1. Fasten your answer papers to section B securely together.
2. Hand in the following separately:
 - Section A (Part I)
 - Section A (Part II)
 - Section B

The number of marks is given in brackets [] at the end of each question or part question.

ANSWER SCHEME

For examiner's Use	
Section A	
1	/ 15
2	/ 10
3	/ 12
4	/ 8
5	/ 12
6	/ 10
7	/ 13
Section B	
8 / 9	/ 20
Total	/ 100

This paper consists of ___ printed pages.

[Turn over]

Section A (Part I)
Answer **all** the questions in this section.

QUESTION 1

Protein function depends on the specific shape of the newly synthesized protein. The tertiary structure of globular proteins depends on the way they are folded.

In the cytosol, chaperone proteins can bind reversibly to newly synthesized polypeptides and shield them from other molecules. This prevents the R groups of amino acids from forming bonds with the wrong molecular partners, hence allowing the polypeptides to fold correctly.

Fig. 1.1 below shows how chaperone proteins work.

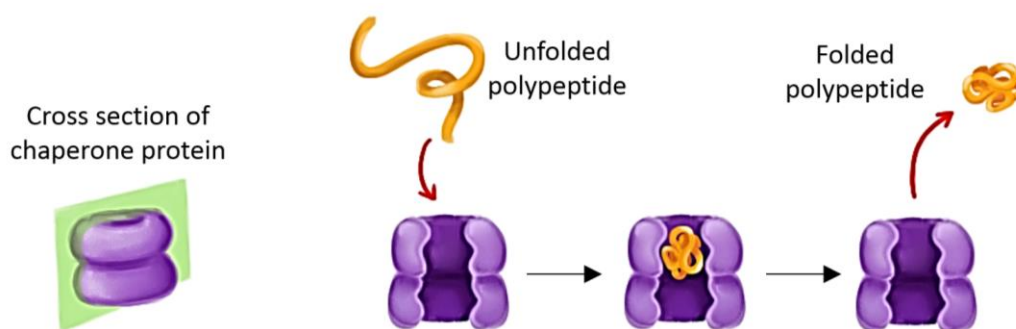


Fig. 1.1

- a) With reference to Fig. 1.1, explain how chaperone proteins regulate gene expression. [2]

[Information from figure]

- **Idea that** Chaperone proteins allow unfolded polypeptides to enter, provide them with an isolated environment to fold, and then allow the folded polypeptides to exit.

[Explain]

- It regulates gene expression at the post-translational level, allows production of functional proteins.

- b) Newly formed polypeptides, together with their chaperones, must be moved from the site of synthesis to their target location in the cell, where they adopt their final shape to carry out their specific function.

Suggest one reason why this movement is unlikely to be by diffusion. [1]

- Protein molecules attached to chaperones are large, hence the movement will be too slow over large distances / **AVP**
- They move to specific locations in a cell (e.g. Golgi apparatus or cell surface membrane), while diffusion is directional / depends on a concentration gradient / **OWTTE**.

When a protein fails to fold correctly upon synthesis or misfolds at a later stage in its cellular life time, it can no longer fulfil its biological function. This may happen as a result of either:

- Ageing, which is associated with a decrease in expression of chaperone proteins, or
- An inherited mutation in a gene, which causes a change in the amino acid sequence of a protein.

Protein misfolding can result in diseases because some misfolded proteins tend to aggregate into fibrils, which may result in cell death.

Fig. 1.2 shows how misfolded proteins aggregate.

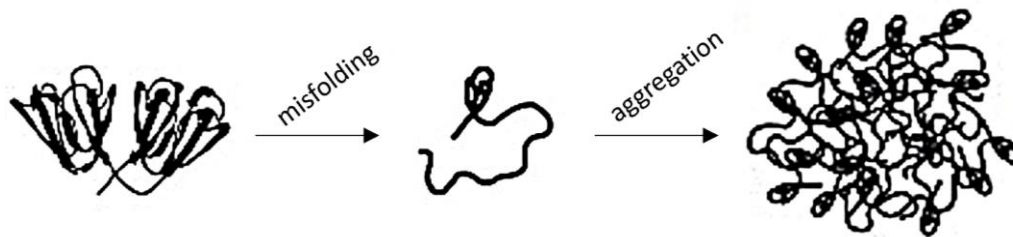


Fig. 1.2

c) Describe how cytosolic proteins are broken down in cells. [2]

- Proteins to be degraded are tagged with ubiquitin proteins.
- Large protein complexes called proteasome recognise the ubiquitin and degrade the tagged cytosolic proteins.
- Hydrolysis of peptide bonds to produce smaller peptides / amino acids by proteases / peptidases.

[Reject: Degradation path for membrane proteins → ubiquitination triggering endocytosis and the subsequent fusion of the endosome with lysosomes.]

d) With reference to Fig. 1.2, suggest why misfolded proteins tend to aggregate in the cell. [2]

- **Idea that** hydrophobic amino acids are exposed, hence misfolded proteins may be insoluble in water.
- Hydrophobic interactions between misfolded proteins form aggregate
- **AVP [max 1]**
 - Misfolded proteins may not have a shape that is complementary to the active site of protease.
 - Change in folding of proteins may cause its binding site to proteases to be inaccessible.
 - Fibrils are too large to enter proteasome to be degraded.

An example of a misfolding disease is Alzheimer's disease, a neurodegenerative disease. In Alzheimer's disease, there is an accumulation of misfolded amyloid β protein in the neurons of patients, leading to symptoms of the disease.

Hereditary forms of Alzheimer's disease have been associated with a mutation in the amyloid β precursor protein (APP) gene, which encodes the APP protein. The APP protein is a transmembrane glycoprotein found on the surface of many cells, such as neurons. The proteolysis of the APP by enzymes produces the amyloid β protein.

e) Describe how the APP protein on the cell surface membrane is formed after its polypeptide chain has been synthesized. [4]

- The APP polypeptide is embedded/anchored on the endoplasmic reticulum membrane...
- ...and the polypeptide chain folds into its 3D conformation in the lumen.
- In the ER/GA lumen, the APP undergoes glycosylation.
- ER vesicles with the embedded APP buds off from the ER and travel towards and fuses with the cis face of the Golgi apparatus.
- The protein gets embedded in the GA membrane, and the modified protein is packaged into a Golgi vesicle which buds off from the GA.
- [compulsory] The Golgi vesicle with the embedded protein moves towards and fuses with the cell surface membrane via exocytosis.

One mutation on the APP gene involves one codon being changed from GCC to GUC. Fig. 1.3 shows the genetic code.

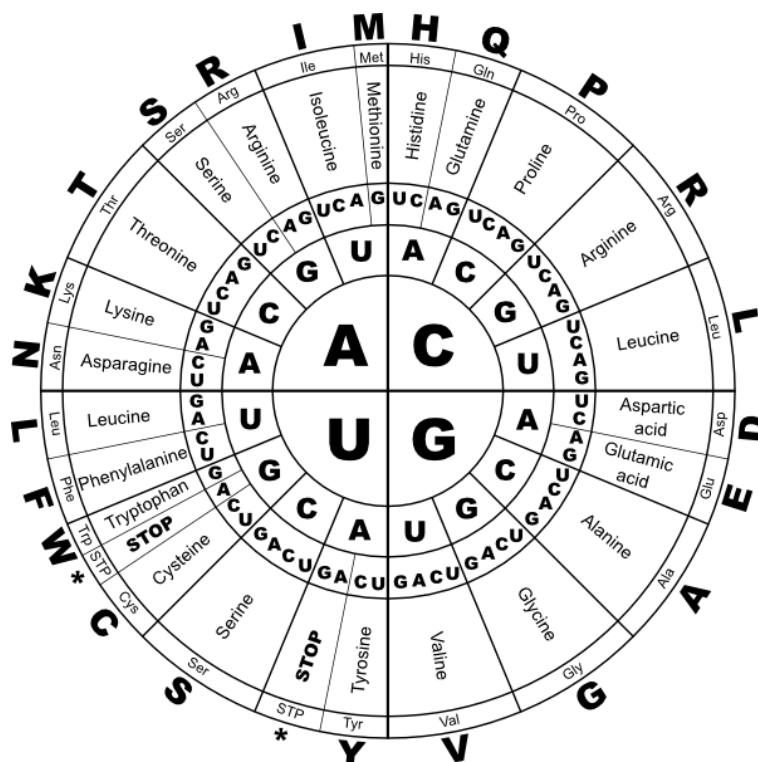


Fig. 1.3

f) Explain the term *codon*. [2]

- A codon is three nucleotides/bases on mRNA that specify an amino acid.
- Stop codons/UAA, UAG, UGA signal the termination of translation.
- It complementary base-pairs with the corresponding tRNA anticodon / DNA triplet.
- It is transcribed from DNA triplet on template strand.

g) With reference to Fig. 1.3, describe the mutation that has occurred. [2]

- Single base pair substitution mutation from guanine to adenine on DNA template strand.
[Reject: Mutation on mRNA]
- Results in the substitution/replacement of an amino acid from alanine to valine.

[Total: 15]

QUESTION 2

In normal cells, the cell cycle is controlled at checkpoints. These checkpoints involve proteins called Cyclin-Dependent-Kinases (CDKs). CDKs are inactive, but can be activated by binding to cyclins to form cyclin-CDK complexes. Active CDKs activate target proteins that lead to cell cycle progression.

Different cyclin-CDK complexes trigger different steps in the cell cycle. Fig. 2.1 shows how the formation of one type of cyclin-CDK complex triggers mitosis.

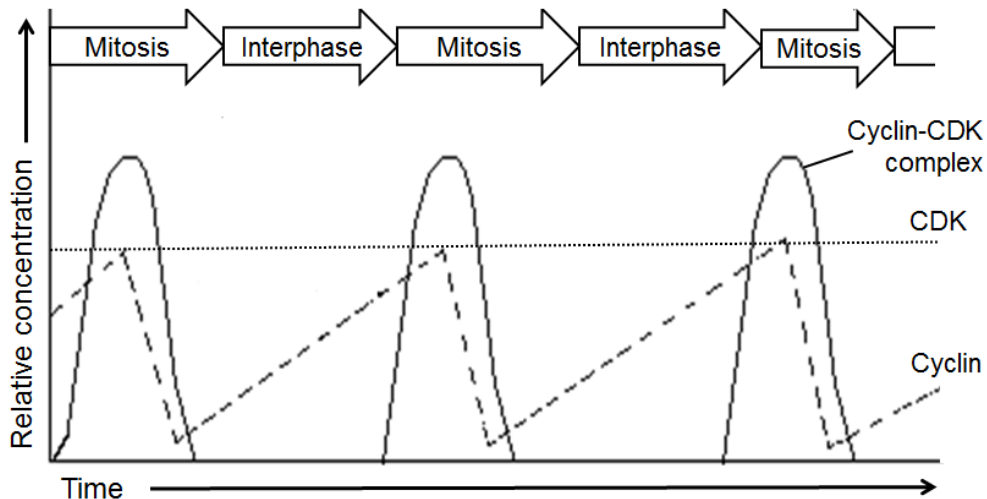


Fig. 2.1

a) With reference to Fig. 2.1, explain how the levels of cyclin-CDK complexes fluctuate. [2]

- Concentration of CDK remains constant, while concentration of cyclin fluctuates.
- Cyclin concentration increases due to synthesis, and decreases due to degradation / **OWTTE**
- As cyclin concentration increases, more cyclin bind to CDK, hence concentration of cyclin-CDK complexes increase during mitosis.

b) Describe how target proteins can be activated by active CDKs. [2]

- CDKs phosphorylate / add phosphate group to target proteins using ATP.
- Target proteins change conformation and is activated.

OR

Allows other proteins to bind to the phosphorylated target proteins.

c) Describe the roles of the checkpoints that control the cell cycle. [3]

- G₁ checkpoint ensures sufficient nutrients/growth of the cell for DNA replication and cell division.
- G₂ checkpoint checks for DNA damage/mutation.
- M checkpoint checks for the proper attachment of spindle fibres to kinetochores/centromeres of chromosomes.

d) Explain how the dysregulation of checkpoints may lead to cancer. [3]

- Gene/chromosomal mutations are passed on to daughter cells...
- ...as the cell cycle is not halted to repair damage / DNA damage not repaired.
- This results in the daughter cell having gain-of-function mutation of proto-oncogenes (e.g. Ras) and/or loss-of-function mutation of tumour suppressor genes (e.g. p53).
- Accumulation of these mutations in a single cell would then lead to uncontrolled cell division, forming a primary tumour.

[Total: 10]

QUESTION 3

Fig. 3.1 shows the methylation pattern of a segment of DNA from a human chromosome taken from three different cell types.

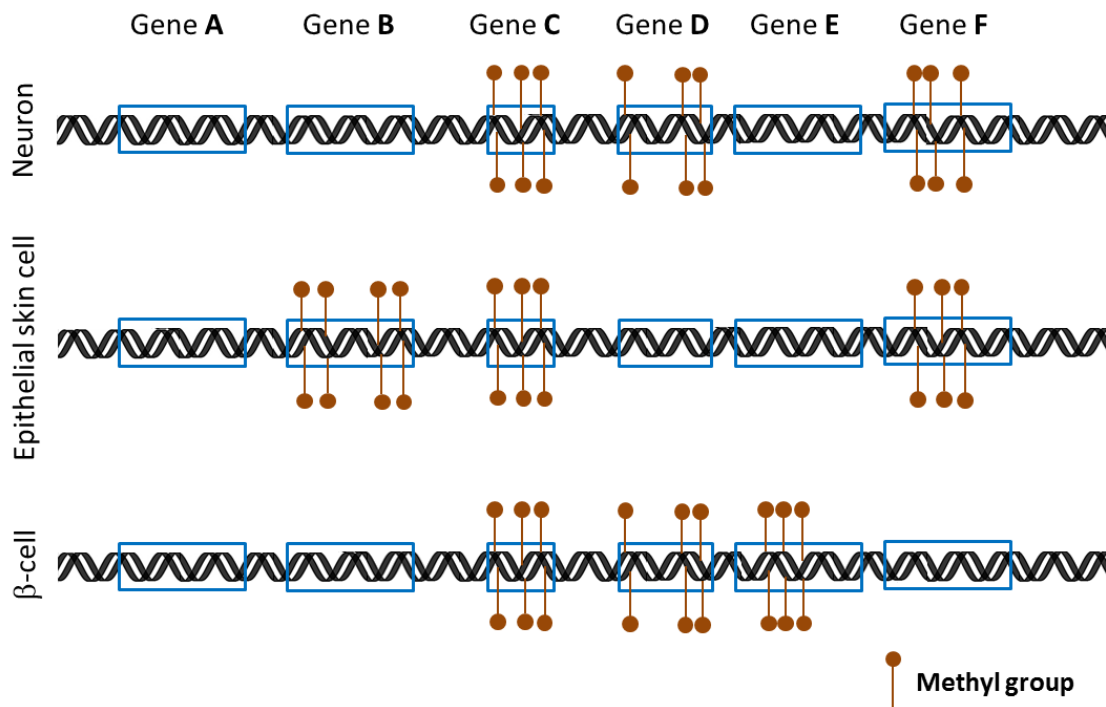


Fig. 3.1

a) Explain the effect of DNA methylation on gene expression. [4]

- Methylated DNA recruits histone deacetylase
- Removal of acetyl group from lysine and arginine residues on histone tail, restores positive charge to lysine and arginine residue
- Ionic bonds form between positively-charged histone tails and negatively charged DNA, leading to compaction of chromatin
- Transcription factors and RNA polymerase cannot access the promoter of gene, hence prevent gene expression
- Long-term inactivation of genes

b) Explain why the pattern of DNA methylation differs in the three cell types. [3]

- All somatic cells contain the same set of genes
- Differential / spatial gene expression
- **Idea that** cell-type specific genes are only expressed in that specific cell (e.g. gene E expressed only in β-cell), hence are not methylated. / **reverse argument**
- **AVP [e.g. elaboration of point 3]**

[Reject: save resources. The main idea is on differential gene expression]

- c) Suggest why gene **A** is not methylated in all the cell types and hence suggest a possible identity of gene **A**. [2]
- **Idea that** gene A is a housekeeping gene that is required for normal functioning of all cell types
 - E.g. Gene that codes for RNA polymerase / aminoacyl tRNA synthetase / glycolytic enzymes / **AVP**

Methylation of DNA always occurs on the cytosine base of a CG dinucleotide. The methylation pattern is heritable when a cell divides by mitosis. Fig. 3.2 illustrates this process.

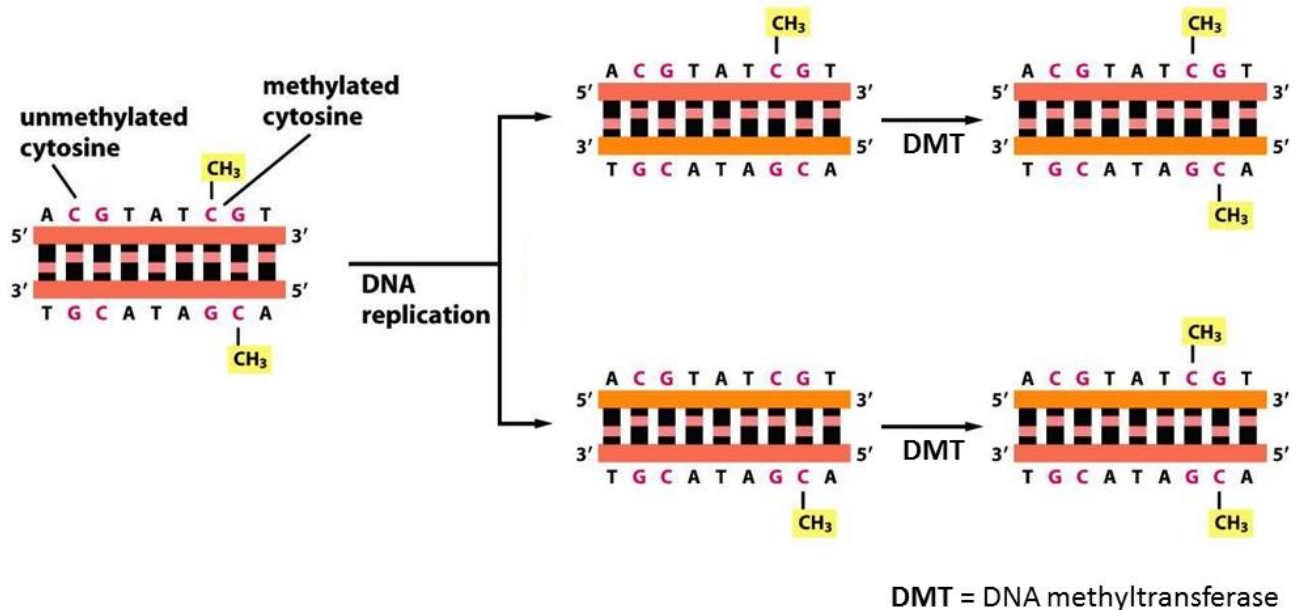


Fig. 3.2

- d) Using information in Fig. 3.2, suggest how it is possible for the methylation pattern to be inherited. [3]
- Active site of DMT is complementary in shape [mark once] and binds to the methylated cytosine of CG on the parental DNA strand
 - **Idea that** the parental strand with methylated cytosine of CG serves as a signal for DMT to methylate the cytosine of CG on the newly-synthesized strand.
 - Unmethylated cytosine of a CG does not bind DMT and hence is not methylated.

[Total: 12]

QUESTION 4

Blood donated to blood banks undergo stringent HIV testing before being made available to hospitals. Nucleic acid tests have been developed to detect RNA in the HIV particles in the blood plasma and to detect HIV DNA in infected blood cells.

Fig. 4.1 shows the levels of HIV RNA and HIV DNA in the first 50 days of infection.

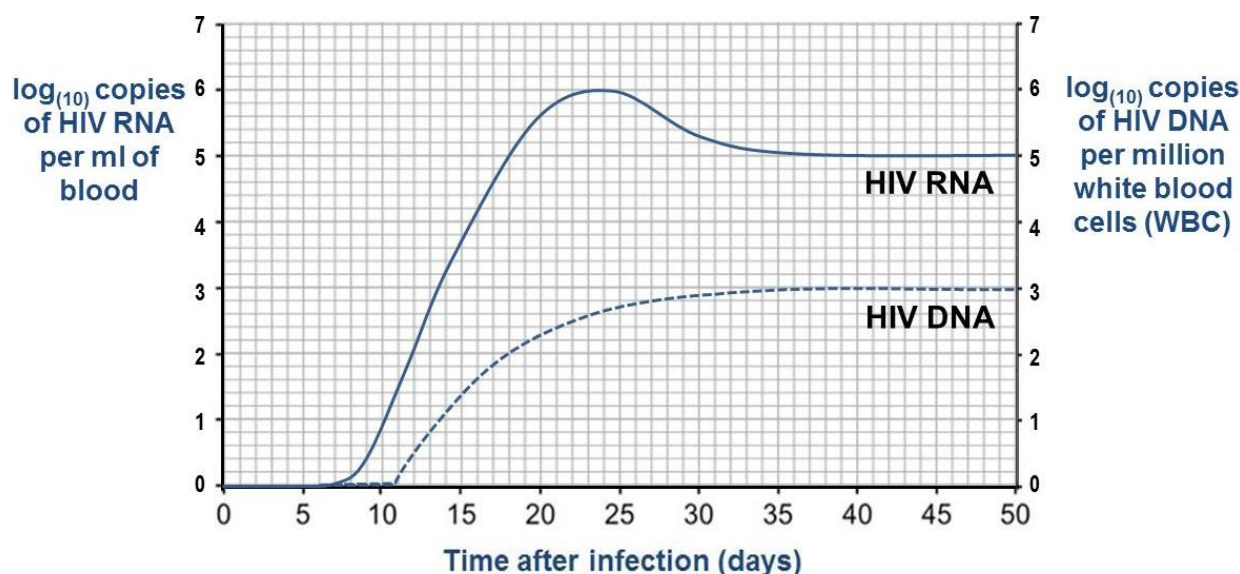


Fig. 4.1

a) State a specific type of white blood cell from which the levels of HIV DNA can be measured. [1]

- $CD4^+$ T cells / Helper T cells / macrophages

b) With reference to Fig. 4.1,

i) Explain why HIV is considered to be a retrovirus. [3]

- HIV carries a **(+) single-stranded RNA** which acts as a template for reverse transcription into DNA in the host cell...
- ...by **reverse transcriptase** which forms double stranded **cDNA**
- Thus genetic information flows in the reverse direction, from RNA to DNA

[compulsory]

- **Idea that** HIV DNA only increases after HIV RNA increases in copies in the blood +

Cite relevant data: e.g. From day 6/7 to day 22 [accept 22-25], the amount of HIV RNA increases from 0 $\log_{10}$ / 1 to 6 $\log_{10}$ / 1,000,000 copies per ml, while only from 11 days after infection does the level of HIV DNA increase from 0 $\log_{10}$ to 3 $\log_{10}$ / 1000 per million WBC.

ii) Account for the levels of HIV DNA after 35 days of infection. [3]

- From day 35 to 50 / beyond 35 days, the levels of HIV DNA **remains constant at 3.0 $\log_{10}$ copies per million WBC.**
- This is because HIV cDNA is **integrated** as a **provirus** into the helper T cell genome and remains latent
- $CD4^+$ T cells are **not dividing**, hence amount of DNA remains constant
- In addition, no new virus released to infect other healthy WBC

- c) Antibiotics are often given to HIV/AIDS patients to treat secondary bacterial infections, but not to treat the HIV infection.

Suggest why.

[1]

- Viruses do not have ribosomes / cell walls / cell membranes/ cellular machinery/AVP, which are targeted by antibiotics, unlike in bacteria
- **AVP**

[Total: 8]

QUESTION 5

The mode of inheritance for shape of summer squash can be investigated by a cross between two summer squash plants with disc-shaped fruit. The cross resulted in offspring with 3 different phenotypes in the following numbers:

Disc-shaped fruit	1071
Sphere-shaped fruit	720
Long fruit	123

Shape of summer squash is determined by 2 gene loci, **A/a** and **B/b**.

a) Explain how different fruit shapes in summer squashes come about. [3]

- The fruit shapes in summer squashes are controlled by **epistasis**, where there is duplicate genes with cumulative effect. **[Reject: recessive/dominant epistasis]**
- Presence of a dominant allele from either gene / genotype A_bb or $aaB_$ codes for an enzyme that produces sphere-shaped fruit from long fruit.
- If dominant alleles from both genes are present / genotype $A_B_$, the fruit is further flattened into disc-shape.
- Absence of any dominant alleles / homozygous recessive plant do not produce any functional enzyme, and hence resulted in long fruits.

[Any 3]

[Penalise once if no reference to enzyme]

b) With this understanding of their mode of inheritance, a breeder crossed two sphere-shaped summer squash plant together. None of the offspring produced gives sphere-shaped fruit.

Construct a genetic diagram to explain the cross. [3]

Parental phenotype Sphere-shaped x Sphere-shaped

Parental genotype $AAbb$ x $aaBB$

[1m – correct parental genotypes]

Gametes

Ab

x

aB

[1m – correct gametes (circled)]

F_1 genotype

$AaBb$

F_1 phenotype

All Disc-shaped fruit

[1m – correct F_1 genotypes AND phenotypic ratio]

c) A test cross is used to determine the genotype of an organism.

Comment on the effectiveness of using a test cross to determine the genotype of a disc-shaped squash. [3]

- **Cross** the disc-shaped squash plant with unknown genotype with a **long-shaped** squash plant that is **homozygous recessive / aabb**. [1]

A test cross is only effective in determining the genotype when:

- the disc-shaped squash plant is **homozygous at both gene loci / AABB**, **all** the offspring will have **disc-shaped** fruit.
- the disc-shaped squash plant is **heterozygous at both loci / AaBb**, the offspring will have **all 3 phenotypes**.

Not able to differentiate:

- between plant that is **heterozygous at either one loci / AaBB or AABb**, the offspring will have the phenotype of **disc-shaped** fruit and **sphere-shaped** fruit.

[Any 2 of the 3 possible results]

Squashes also possess different fruit colours. Fruit colours of squash are determined by two other gene loci on 2 different chromosomes. The colours of squash are determined by gene locus **C/c**. The presence of a dominant allele, **C** results in white squash, while the recessive allele gives rise to coloured squash. At the second gene locus **G/g**, allele that codes for yellow is dominant to allele that codes for green.

A cross between two heterozygous white squash plants gives phenotypes in the following numbers:

White squash	250
Yellow squash	60
Green squash	10

The expected ratio for the above cross is 12:3:1.

The χ^2 distribution table and equation to calculate χ^2 is shown below.

number of degrees of freedom (v)	probability
	0.05
1	3.84
2	5.99
3	7.82
4	9.49

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

O = observed result
E = expected result

d) Using the equation and χ^2 distribution table given above, explain if epistasis is the correct explanation for the results obtained. [3]

$$\chi^2 = \frac{(240-250)^2}{250} + \frac{(60-60)^2}{60} + \frac{(10-20)^2}{20} = 0.42 + 0 + 5 = 5.42$$

- Calculated $\chi^2 = 5.42$.
- At degree of freedom = 2, calculated χ^2 is **less than the critical value of 5.99**.
- The **probability** that the difference between expected and observed number is due to chance is **more than 0.05 / p > 0.05**.
- Difference is **not significant / due to chance**, the observed number **conform to the epistatic ratio 12:3:1**, hence epistasis is the correct explanation.

[Total: 12]



Section A (Part II)

Answer **all** the questions in this section.

QUESTION 6

Contraction of airway smooth muscle is controlled by the release of acetylcholine from presynaptic neurones. Binding of acetylcholine to G-protein coupled receptors (GPCR) embedded in the smooth muscle membrane eventually result in the contraction and growth of smooth muscles. Fig. 6.1 below show part of this signalling pathway.

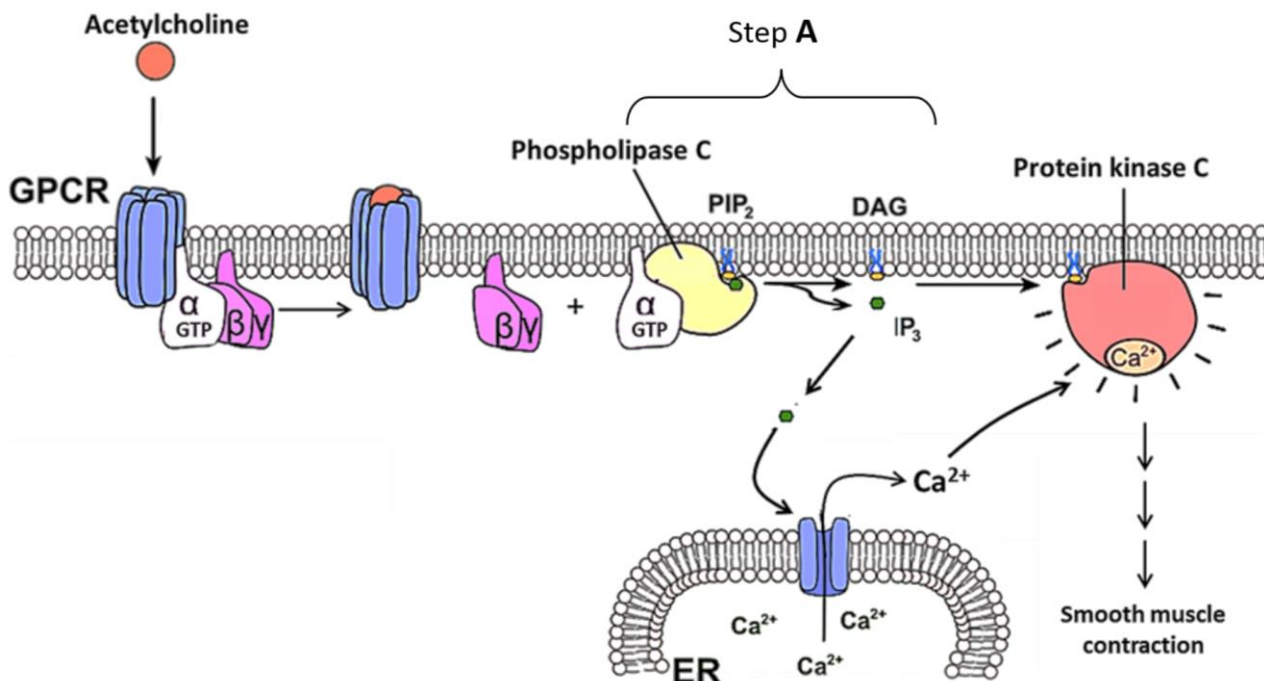


Fig. 6.1

a) With reference to Fig. 6.1,

i) describe one distinctive structural feature of the G-protein coupled receptor. [1]

- Consist of seven transmembrane domains / α -helices spanning the membrane.
[Reject: Link to G-protein]

ii) explain the role of G-protein coupled receptor. [2]

- Consist of a specific receptor site that has a shape complementary to that of acetylcholine.
- When bound, it undergoes conformational change and activates G-protein.

iii) explain the significance of step **A** in facilitating an effective transduction. [3]

- Activated phospholipase C catalyses the cleavage of many PIP₂ to form (many) second messengers – DAG and IP₃. *(idea of 'many' PIP₂ cleaved or 'many' DAG and IP₃ produced)*
- Thus amplifying the signal / a single ligand can produce a large cellular response
- IP₃ is water-soluble and can diffuse throughout the cell/cytosol to activate IP₃-gated Ca²⁺ channels.

OR

DAG is lipid-soluble and can diffuse within the plasma membrane to activate protein kinase C.

b) Describe **one** difference in the roles of Ca²⁺ in triggering contraction of smooth muscles and in the release of acetylcholine in neurons. [2]

- In smooth muscles, Ca²⁺ act as second messenger which binds to and trigger conformational change in PKC, activating it.
- While Ca²⁺ in synaptic knob triggers the vesicles containing acetylcholine to fuse with the presynaptic membrane, to release the acetylcholine into the synaptic cleft via exocytosis.

c) A mutation resulted in high concentration of DAG and IP₃ being produced in the muscle cell.

Assuming that the phospholipase C functions normally, explain how a mutation can lead to the observation above. [2]

- Gain of function mutation in the gene which codes of GPCR / promoter / control elements
- Result in hyperactive receptor which binds to acetylcholine permanently/continuously to trigger the signal pathway / higher amount of proteins

OR

- Gain of function mutation in the gene which codes of GPCR / promoter
- Result in hyperactive receptor which binds to and activates G-protein permanently/continuously to trigger the signal pathway / higher amount of proteins

OR

- Gain of function mutation in the gene which codes of G-protein.
- Result in hyperactive G-proteins which binds to and activates PLC permanently.

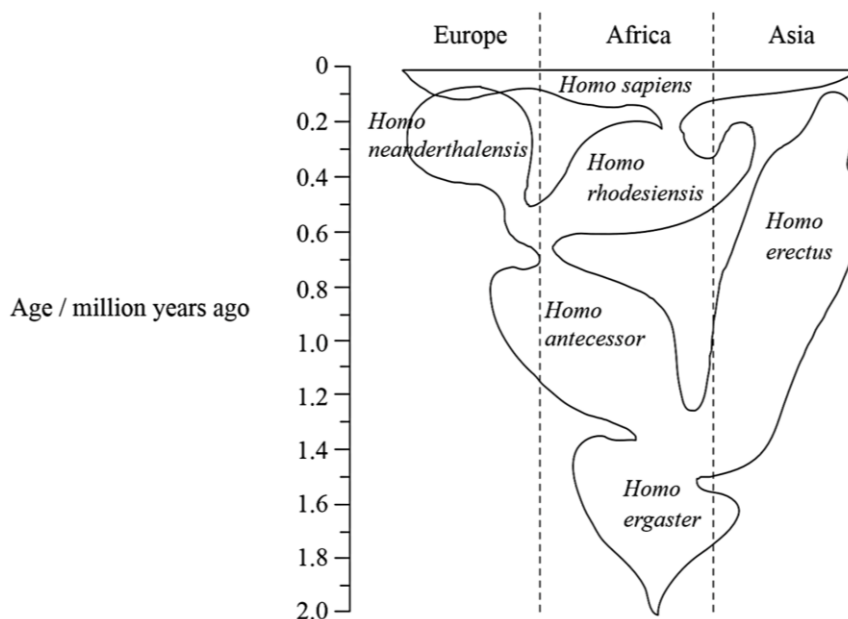
[Total: 10]

QUESTION 7

a) Outline the role of variation in evolution. [4]

- Members of a population are genetically different thus exhibit phenotypic variation within the population / **idea of** different alleles in a population
- Some variations allow an individual to be better adapted / at a selective advantage
- Better adapted individuals survive to reproductive age to pass on advantageous alleles to their offspring.
- Individuals with less adaptable variation die before reproductive age.
- **Idea that** allele frequency changes over time.
- Variation is important as it decreases the chances of extinction as a result of natural selection.

b) Recently discovered fossilized partial skulls from Ethiopia have raised new questions about early human evolution. This has led to different theories about the origins of *Homo sapiens*. One of these theories is illustrated in Fig. 7.1.



[Source: Stringer, Nature (2003), 423, pages 692-695]

Fig. 7.1

i) Identify the species that shows the greatest geographical distribution. [1]

- *Homo sapiens*

ii) Suggest, with reason, whether *H. antecessor* or *H. erectus* would likely provide more fossil evidence. [2]

- *H. erectus* as it existed longer,
- **[Quantity]** Thus greater number of individuals, hence more fossils can be formed.

OR

- *H. erectus* as it existed more recently
- **[Quality]** Hence fossils are better preserved / more complete / more accessible.

- c) The evolution of groups of living organisms can be studied by comparing the base sequences of their DNA. If a species diverges into two groups, differences in base sequence between the two species accumulate gradually over long periods of time. The number of differences can be used as an evolutionary clock.

Samples of DNA were recently obtained from fossil bones of a Neanderthal (*Homo neanderthalensis*). A section of the DNA from the mitochondrion was chosen for study, as it shows a high level of variation in base sequence between different individuals. A section of the Neanderthal mitochondrial DNA was sequenced and compared with sequences from 994 humans and 16 chimpanzees. The bar chart below shows the number of base sequence differences found among humans, between humans and the Neanderthal, and between humans and chimpanzees.

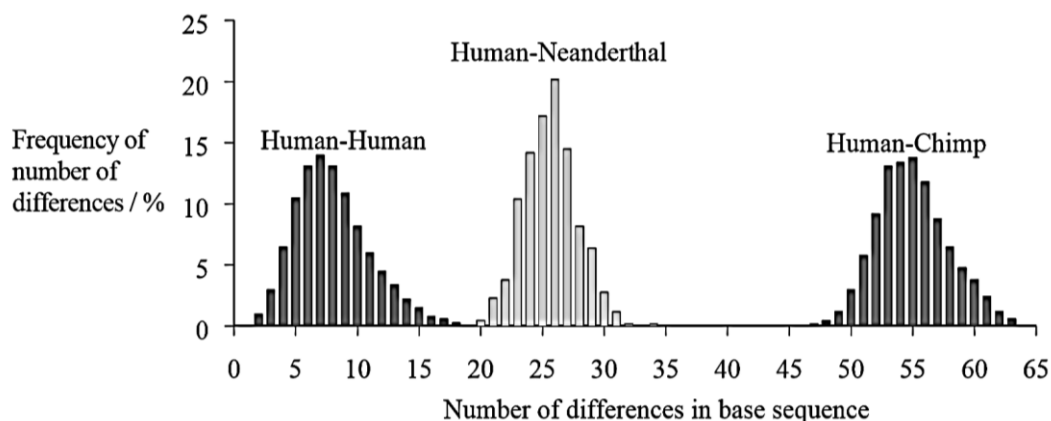


Fig. 7.2

- i) The number of base sequence differences between pairs of humans varied from 1 to 18.

State the number of differences shown by the highest percentage of pairs of humans. [1]

- 7

- ii) Humans and Neanderthals are both classified in the genus *Homo* and chimpanzees are classified in the genus *Pan*.

Discuss whether this classification is supported by the data in Fig. 7.2. [3]

- **Supported.** Fewer nucleotide differences between humans and Neanderthal (20 to 34) than humans and chimps (47 to 63).
- Humans and Neanderthal **share a more recent common ancestor** than human and chimp / diverged more recently from a common ancestor / more closely related.
- Lesser time to accumulate mutation / genetic differences.

- d) Explain why the evolutionary clock supports the neutral theory of molecular evolution. [2]

- Molecular clock shows a **constant rate of (accumulation of) mutation**.
- Includes mutations to **non-coding sequence**, **silent mutation** to coding sequence, and mutations that have **no effect on phenotype**.
- These mutations accumulate as they are **selectively neutral** / not selected for or against.

[Total: 13]

Section B
Answer **one** question.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labeled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in questions **(a)**, **(b)**, etc., as indicated in the question.

QUESTION 8

a) Compare tropocollagen and amylose. [7]

Similarities [at least 1]

1. Both are **polymers**/ made up of many monomers.
2. Both are macromolecules that are **insoluble in water**.
3. **Length** of chains may **vary**.
4. Both are stabilised by **hydrogen bonds**.

Differences [max 5]

	Tropocollagen	Amylose
6. Elements	C, H, O, N, S	C, H, O
7. Nature	Fibrous protein	Polysaccharide
8. Monomer	Amino acid	α -glucose
9. No. of types of monomers	20 types	1 type
10. Bond between monomer	Peptide bond	α -1,4 glycosidic bonds
11. Hydrogen bonds	Hydrogen bonds between chains	Hydrogen bonds within chain
12. Overall shape	Linear (chains) shape	Helical shape
13. No. of chains	3 polypeptide chains	1 polysaccharide chain
14. Function	Structural material found extensively in connective tissue, bones, cartilage, dermis (skin), blood vessels and teeth	Energy store (in plant cells)
15. Site of synthesis	Rough endoplasmic reticulum and Golgi apparatus	Cytosol
16. AVP	AVP	AVP

b) The optimum pH for the activity of rubisco is pH8 (alkaline).

Explain why the illumination of chloroplasts leads to optimum pH conditions for rubisco. [7]

1. Photolysis of water at PS-II to form proton and electrons
2. To replace the excited electrons emitted from photosystems
3. Transported along the thylakoid membrane
4. By electron carriers of progressively lower energy levels
5. Energy released from electron transport
6. Used to pump protons from stroma into thylakoid space
7. Lesser amount of protons in the stroma raises pH to alkaline
8. RUBISCO is in stroma

c) Describe the differences in behavior of homologous chromosomes during mitosis and the first division of meiosis. [6]

	Mitosis	Meiosis I
1. Synapsis?	• During prophase , no synapsis / no pairing of homologous chromosomes.	• During prophase I , synapsis occurs/ homologous chromosomes pair up to form bivalents .
2. Crossing over?	• During prophase , no crossing over occurs between homologous chromosomes.	• During prophase I , crossing over occurs between non-sister chromatids of homologous chromosomes at chiasmata.
3. Arrangement at equator?	• During metaphase , all the individual chromosomes arrange in a single row on the equator of the cell.	• During metaphase I , homologous chromosomes arrange in two rows on the equator of the cell.
4. Independent assortment?	• During metaphase , no independent assortment occurs since sister chromatids are identical.	• During metaphase I , independent assortment of chromosomes occur.
5. Centromere divides?	• During anaphase , centromere of each chromosome divides .	• During anaphase I , centromeres do not divide .
6. Identical sister chromatids?	• During anaphase , sister chromatids of each chromosome separate . OR During anaphase, separating chromatids are identical.	• During anaphase I , homologous chromosomes separate . OR During anaphase I, separating chromosomes are not identical.
7. AVP	• AVP	• AVP

[Note if students answered in terms of just 'chromosomes' rather than 'homologous chromosomes. Exercise **flexibility** during marking for point 3-6]

QUESTION 9

- a) The cell surface membrane consists of phospholipids and cholesterol. Describe how their structures and properties are related their functions. [8]

PHOSPHOLIPIDS	
Structure and property [max 4]	Function [max 4]
1. Amphipathic - comprises hydrophobic fatty acid tails and hydrophilic phosphate heads. 2. Forming a lipid bilayer where hydrophobic fatty acid tails face inwards and hydrophilic phosphate heads face outwards/the aqueous cytosol and extracellular matrix. 3. Phospholipids are held by weak hydrophobic and hydrophilic interactions, hence can move laterally/are fluid. 4. Carbohydrate chains can be linked to lipid to form glycolipids. [Reject: Consists of one glycerol, two fatty acid chains and one phosphoric acid group – no corresponding function].	2a. Lipid bilayer separates cell from the extracellular environment. 2b. Hydrophobic region allows it to act as a barrier/regulates transport across membrane, as it allows only small, hydrophobic/lipid-soluble/non-polar substances to pass through. [Max 2] 3a. Idea that it can form vesicles, OR 3b. Its fluidity create transient gaps to allow diffusion of small, non-polar molecules, OR 3c. Its fluidity allows the membrane to can reseal itself if it is disrupted, OR 3d. AVP 4a. Glycolipids are involved in cell-cell recognition. Carbohydrate chains serve as cell surface marker that is unique in different tissues of an individual, and between individuals [mark once].
CHOLESTEROL	
5. Amphipathic – hydrophilic OH group and hydrophobic hydrocarbon rings. 6. Is embedded within the membrane via hydrophobic and hydrophilic interactions with phospholipids.	6a. Regulates fluidity of the membrane. 6b. At high temperatures, it restricts movement of phospholipids, thus prevents the membrane from becoming too fluid. 6c. At low temperatures, it prevents close packing of the phospholipids, thus prevents the membrane from becoming too rigid.

b) Describe what happens to pyruvate if yeast is deprived of oxygen. [5]

1. When deprived of oxygen, yeast undergoes anaerobic respiration, where only glycolysis and alcoholic fermentation take place.
2. Pyruvate produced during glycolysis does not undergo Link reaction / Krebs cycle.
3. Instead, pyruvate remains in the cytosol/cytoplasm, ...
4. ...where it is decarboxylated to form ethanal, where CO₂ is released.
5. Ethanal is then reduced to ethanol by NADH,...
6. which is in turn oxidised to regenerate NAD for glycolysis to continue.
7. Alcoholic fermentation is an irreversible reaction catalysed by alcoholic dehydrogenase.

c) Explain the meaning of the term homeostasis with specific reference to the control of raised blood glucose concentration in mammals. [7]

[Definition]

1. Homeostasis is the maintenance of a constant internal environment despite changes in the external environment.
2. Two basic principles govern homeostatic mechanisms: self-regulation and negative feedback.
3. In self-regulation, the control mechanism is triggered by the parameter that is to be regulated.
4. In negative feedback, the mechanisms restore the stable internal environment back to set point, by moving the parameter in the opposite direction of the change.

[Homeostatic control w.r.t. to raised blood glucose]

5. Set point for blood glucose level is 90mg glucose/100cm³.
6. (self-regulation) The rise in blood glucose level from set point triggers the control mechanism to bring it back down to set point.
7. The rise in blood glucose level is detected by the alpha and beta cells of the Islets of Langerhans in the pancreas (detector).
8. Beta cells are triggered to release insulin into bloodstream, while alpha cells stop glucagon secretion.
9. Insulin travels via bloodstream and acts on liver and muscle cells (effectors), which bring about responses that restore the set point. [1]
 - For eg. Stimulates fusion of vesicles containing glucose-carrier/glucose-transporter proteins with the plasma membrane so as to increase rate of uptake of glucose. [1]
 - AVP [1]
10. These cellular responses result in a decrease in blood glucose level back to the set point of 90mg glucose/100cm³ of blood. This is negative feedback regulation of blood glucose level.

[Total: 20]